

Measurement of Terahertz Band Water Vapor Absorption Spectrum for the Realization of High-Precision Optical Humidity Measurement

Nihon University, College of Science and Technology, Dept. of Electronics, 4th Year
Tomoki Sumiyama

1. Research Background

Humidity is a familiar physical quantity, even to elementary school students. Recently, the relationship between temperature and humidity has gained attention in relation to illnesses like viral infections and influenza. Humidity is also considered crucial in various fields, such as managing low-humidity environments in battery manufacturing for electric vehicles as part of the move toward "decarbonization," and its effects on the quality and efficacy of pharmaceuticals due to the transition of crystal polymorphs into hydrates and anhydrates.

Commercially available hygrometers currently have extremely low accuracy. The humidity accuracy of a typical thermo-hygrometer is $\pm 5\%$, and even high-precision models are only around $\pm 2\%$. Furthermore, errors tend to increase when the object's temperature is outside the measurement range or in low-humidity conditions, leading to accuracy worse than stated above. High-precision measurement of trace gases in the atmosphere is performed using non-dispersive infrared absorption (NDIR), an optical measurement method. However, it is not yet practical for water vapor, which has strong absorption in the mid-infrared region.

In recent years, terahertz waves—electromagnetic waves with frequencies from 100 GHz to 10 THz (corresponding to wavelengths from 1 mm to 0.01 mm) located between light and radio waves—have been gaining attention. Terahertz spectroscopy

using these waves can detect intermolecular vibrations and crystal lattice motions (phonon modes) in solids. This allows for the observation of the entire molecule's vibration, unlike in the infrared band where identifying polymorphs of water and alcohol by observing the O-H vibration of water molecules can be insufficient. Furthermore, several independent absorption lines unique to water vapor exist across the broad spectrum of the terahertz band. The goal is to achieve high-precision absolute humidity measurement over a wide dynamic range by selecting multiple of these absorption lines and measuring the transmitted terahertz intensity.

2. Purpose

To realize an optical humidity measurement system with an accuracy more than an order of magnitude better than existing hygrometers, which have an accuracy of $\pm 5\%$ (or $\pm 2\%$ for high-precision models), by using terahertz spectroscopy to measure and evaluate the water vapor absorption spectrum.

3. Principle

A Michelson interferometer was used for the water vapor spectrum measurement. By measuring the intensity of the transmitted light at various points as the movable mirror moves, an interferogram is obtained. A spectrum can then be derived by applying a Fourier transform to this interferogram.

$I_{sam}(t)$ be the interferogram with the measurement sample present and $I_{ref}(t)$ be

the interferogram without the sample (background), the transmittance T and absorbance A can be expressed by the following equations

Transmittance

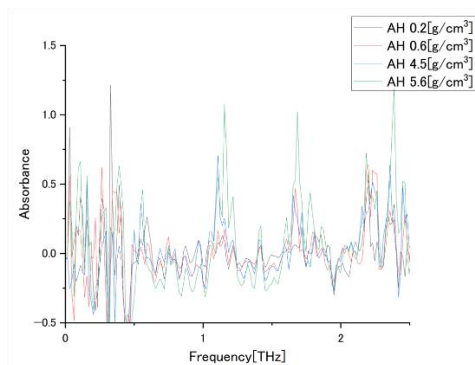
$$T = \frac{\int I_{sam}(t)e^{-j\omega t} dt}{\int I_{ref}(t)e^{-j\omega t} dt} \quad (1)$$

Absorbance

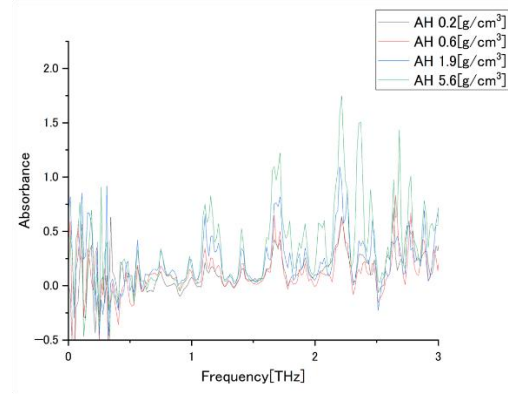
$$A = \log \frac{1}{T} = \log \frac{\int I_{ref}(t)e^{-j\omega t} dt}{\int I_{sam}(t)e^{-j\omega t} dt}$$

4. Results

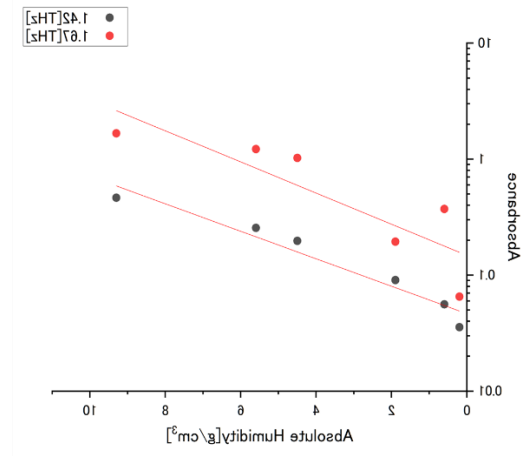
Interferograms were measured for each relative humidity level and for dry air (approx. 0.2%) using a Michelson interferometer. The absorption spectra were then calculated by performing a Fourier transform using OriginPro (Light Stone).



Graph 1: Water vapor absorption spectrum (measured July 26, 2023).



Graph 2: Water vapor absorption spectrum (measured July 6, 2024)



Graph 3: Regression line by least squares method

The measurable range was approximately 0.5 to 2.0 THz. Absorption peaks observed in both measurements were at 0.75, 1.10, 1.15, 1.21, 1.42, and 1.67 THz. These peak values are considered to be the intrinsic absorption spectra of water vapor. Peaks observed only once in either the past or current measurements cannot be definitively identified as intrinsic water vapor peaks. These discrepancies are thought to be due to the quality of the beam (light source) and the

influence of the optical path difference from adjustments to the optical system.

5. Discussion

After calculating the absolute error between the regression line and the measured values, it was found that among the six absorption peaks identified from the measurements, the one at 1.42 THz could be measured with a standard deviation of about 0.04. This means that regarding the objective of precision, there is an error of about 4% from the regression line derived by the least squares method. While a high precision of about 0.1% was achieved at an absolute humidity of 0.6 g/cm³, an error of 4% or more occurred at absolute humidities of 4.5 g/cm³ and higher. It is believed that increasing the number of humidity points for measurement could improve the accuracy of the regression line, thereby reducing the error.

Through this measurement, we believe that the absorbance at 1.42 THz is crucial for the realization of a high-precision optical humidity meter.

授業内で 4 つの AI を比較して、Gemini Pro が 1 番翻訳として、同じような構文を使いまわしてなかつり、読んでいて自然と感じたので、Gemini Pro を用いて、簡易的な論文を作成しました。